

Franziska Hoerbst

Research	Science communication	Art & Activism
1998, Austria	franziska@hoerbst.net	franziskaaah.github.io

Research & education

2024-2025	John Innes Centre, Norwich Research Park, UK
Postdoc	Computational Systems Biology
2020-2024	John Innes Centre & University of East Anglia
PhD	Computational Systems Biology
	Doctoral Thesis supervised by Richard Morris
	“Bayesian Inference in the Analysis of RNA-Seq Data”
2017-2020	University of Vienna, Austria
BSc	Molecular Biology
	Thesis in Computational Biophysics
	Supervised by Bojan Zagrovic at Max Perutz Labs
	“Exploring the Physicochemical Foundation behind Travelling Autologous Protein-mRNA Pairs in the Phloem of Plants”
-2017	Primary and high school, Austria

Science communication, workshops & teaching

2022-2024	UEA Biodiversity and Climate Action Network
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	~Building a movement encompassing students and staff; developing and running workshops and campaigns
2019-2022	5MinutenClimateChance
	Climate science podcast & radio broadcast
	Co-founder, speaker, journalistic researcher & producer
	->Austrian SDG Award 2020
	->Austrian Environmental Journalism Prize 2020
	->Top 5 Salto Awards 2021
2023	Undergraduate student supervision
JIC	Computational and Systems Biology
2022	University teaching assistant
UEA	Mathematics for Computer Scientists
2018-2020	ScienceCenter Network Vienna
	Hands-on science communication
	~Hosting workshops, developing hands-on and tinkering activities for audiences of all ages and backgrounds

Research publications

2022	Exact Bayesian inference for the detection of graft-mobile transcripts from sequencing data
Royal Society Interface	F Hoerbst, M Tomkins, S Gupta, F Apelt, J Kehr, F Kragler, RJ Morris
2023	baymobil: A Python package for detection of graft-mobile mRNA using exact Bayesian inference on RNA-Seq data
Preprint	F Hoerbst, RJ Morris, M Tomkins

2024	A Closed-Form Solution to the 2-Sample Problem for Quantifying Changes in Gene Expression using Bayes Factors
Preprint	F Hoerbst, GS Sidhu, M Tomkins, RJ Morris
2024	Re-analysis of mobile mRNA datasets highlights challenges in the detection of mobile transcripts from short-read RNA-Seq data
Nature Plants	P Paajanen, M Tomkins, F Hoerbst, R Veevers, M Heeney, HR Thomas, F Apelt, E Saplaoura, S Gupta, M Frank, DJ Walther, C Faulkner, J Kehr, F Kragler, RJ Morris
	+ Research briefing: Current RNA-seq evidence of mRNA mobility is largely due to noise in the data
2024	Bayesian Inference in the Analysis of RNA-Seq Data
Thesis	F Hoerbst
2025	On the overinterpretation of mass screening data – the example of mobile mRNA
Trends in Plant Science	F Hoerbst, M Tomkins, RJ Morris
2025	A Bayesian framework for ranking genes based on their statistical evidence for differential expression
Under review	F Hoerbst, GS Sidhu, T Omori, M Tomkins, RJ Morris
2025	What is a differentially expressed gene?
Under review	F Hoerbst, GS Sidhu, M Tomkins, RJ Morris

Further publications & engagement

2024	Let's 'do different': Planetary Citizen Education Proposal
Working paper	E Honeybun-Arnolda, P Stephanides, F Hoerbst, N Longhurst, D Maguire, H Hoechner
2020-2022	5MinutenClimateChance

Podcast	Involvement in writing, editing, or reviewing of 100+ episodes on climate research made accessible and easy to digest
2021	A day trip through phloem
Blog	Easy read research project description
2023	Life out there? Scanning plant space for new communication networks
Blog	Easy read research paper teaser
2023	Activism and Advocacy as a Researcher
Panel	Open discussion at the Annual Assembly of the Tyndall Centre for Climate Change Research

Doing things & being seen

2025	->John Innes Foundation Prize for Excellence in Scientific Research
2025	Movement Research Unit
	~Volunteering to do the research that helps movements win
2024	->Best Presentation Award at MaxEnt24
	Maximum Entropy (MaxEnt) is a historical conferece series in the field of Bayesian statistics, where reserachers from quantum physics and philosophy to economy come together to learn from each other about probability theory and its use
2023	->JIC Director Award
	A yearly award given by Director Graham Moore to individuals to acknowledge and honour outstanding scientific and cultural contributions to the John Innes Centre
2022-2025	UEA Biodiversity and Climate Action Network
	~Particularly involved in the working group on higher education, with the vision of changing

	all curricula towards engaging all student in concepts surrounding global citizenship
2022-2024	JIC Student Voice
	~Running campaigns to support the John Innes student community to thrive
2012-2018	Radius 106.6, Freies Radio Freistadt, Radio Orange 94.0
	Radio presenter, speaker, producer
	->Student award by the Austrian Federal Ministry of Education and OE1 for best broadcast 2016
2017 & 2018	Coordination and execution of charity radio marathons
	4 Tage für den Sozialmarkt, 4 Tage für die Lebenshilfe
2016	->Regional Movie Award @YOUKI Film Festival
2008-2017	->Several accolades for music (We can discuss the concept of music competitions, especially for kids and teenagers)

Finds balance in	Loves speaking	Finds a lot of enthusiasm for
Music and silence	German, English and Spanish	Life
Nature and the outdoors	Python, fluently	The post carbon world
Arts and crafts	At international scientific conferences	Changing climates, environments, structures, hats
Tech and low-tech	At public engagement events (like e.g. Pint of science)	Equality and Justice
Spreading joy and seeing trees upside down	On panels (like previously on Women in STEM, climate activism as researchers)	Colours and Complex Systems

Cooking meals for others	Bayesian Statistics	Information theory
Movement and Movements	With kind humans around a dinner table	Bicycles and Biodiversity
Learning and studying	About importance for community and cultural practices	Slowing down and degrowth

Contact for references

Professor Richard J. Morris	Dr Noel Longhurst	Dr Melissa Tomkins
Group leader at John Innes Centre, PhD supervisor	Lecturer in Energy & Climate Change, School of Environmental Sciences, UEA and founding member of UEA BCAN	Senior Researcher at John Innes Centre, PhD supervisor
Richard.Morris@jic.ac.uk	N.Longhurst@uea.ac.uk	Melissa.Tomkins@jic.ac.uk